

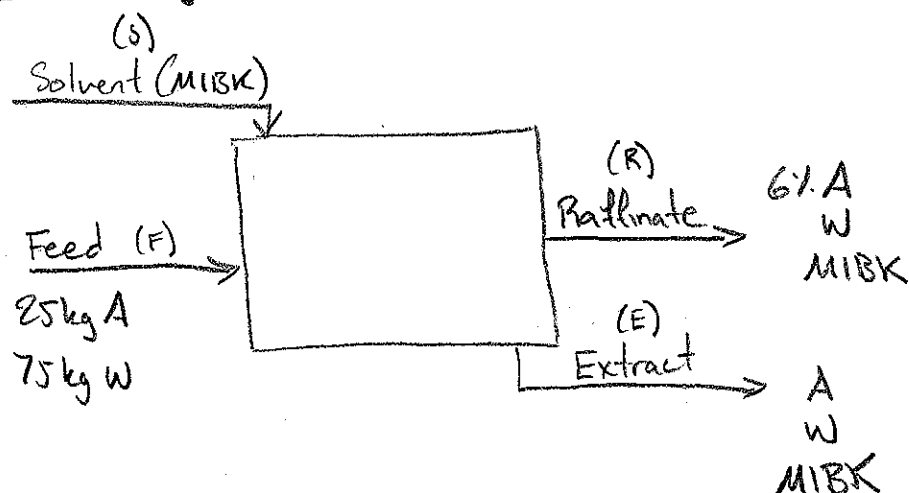
①

Homework #9

SolutionsChE 210, Fall '08
Yergenny Strashnov
(TA)

(1) Murphy, 5.40

Take 100 kg/hr feed as basis...

Flow Diagram:

note:

A ~ acetic acid
W ~ water
M ~ MIBK

Given the fact that the raffinate stream has a specified comp. of 6% A, we can use Fig. 5.15 to read off the rest of the compositions of both the raffinate and the extract streams.

The mixer-settler unit is at equilibrium, so the streams lie on the sides of the bell-shaped curve in the figure. The raffinate lies on the "Water-Rich Phase" line on the left and the extract is on the "MIBK-Rich Phase" line on the right. The two streams are represented by two single points (one on each side) and are connected by an imaginary tie-line that lies in between the two lowest tie-lines that are shown in the figure. Starting with the 6% A value (vertical axis) we can

② read off the following compositions:

Raffinate

Extract

A: 6%

A: ~5%

note: values are in weight %.

MIBK: ~2%

MIBK: ~92%

W: $100 - 6 - 2 \cong 92\%$

W: $100 - 5 - 92 \cong 3\%$

Once this information is obtained, we can proceed with our analysis as usual:

Material Balances:

$$(A) \quad 25 = .06 \dot{m}_R + .05 \dot{m}_E$$

$$(W) \quad 75 = .92 \dot{m}_R + .03 \dot{m}_E$$

$$(M) \quad \dot{m}_S = .02 \dot{m}_R + .92 \dot{m}_E$$

solve
 $\Rightarrow$

$\dot{m}_R = 69 \text{ kg/hr}$
$\dot{m}_E = 387 \text{ kg/hr}$

$$\& \dot{m}_S = 357 \text{ kg/hr}$$

we can check our values using the overall mass balance on the entire system:

$$\dot{m}_S + \dot{m}_F \stackrel{?}{=} \dot{m}_E + \dot{m}_R$$

$$357 + 100 \stackrel{?}{=} 387 + 69$$

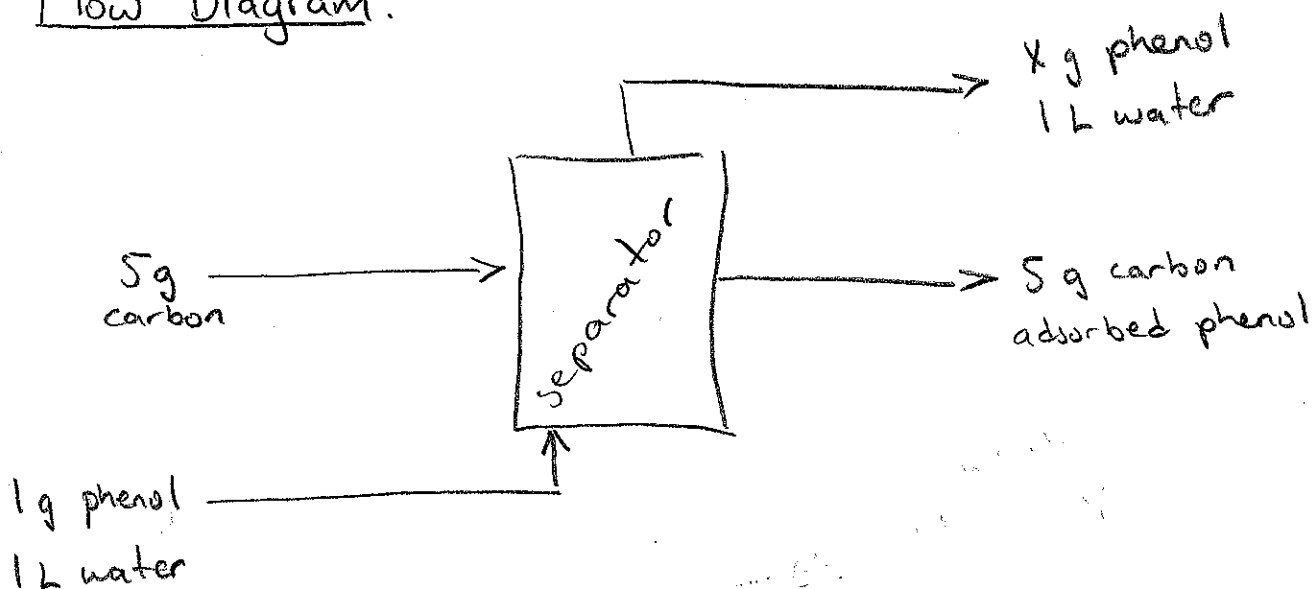
$$457 \stackrel{?}{=} 456 \quad (\text{close enough}) \quad \text{☺}$$

Therefore, we need $\frac{357 \text{ kg solvent}}{100 \text{ kg feed}} = \boxed{3.57 \text{ kg solv. / kg feed}}$

③

(2) Murphy, S.41

Flow Diagram:



We first convert the amt. of phenol to mmol, so that we can use the given diagram.

$$1 \text{ g } \text{C}_6\text{H}_5\text{OH} \cdot \frac{1 \text{ mol } \text{C}_6\text{H}_5\text{OH}}{94 \text{ g } \text{C}_6\text{H}_5\text{OH}} \cdot \frac{1000 \text{ mmol}}{\text{mol}} = 10.638 \text{ mmol phenol}$$

Now, let c_p = concentration of phenol (mmol/L)

& q_p = adsorbed phenol (mmol/g)

Then, the phenol balance is:

$$10.638 = (1\text{L})c_p + (5\text{g})q_p$$

We now use the trial & error method to finish solving the problem by picking points on the curve, reading off their coordinates,

④

and plugging them into our phenol balance equation to check if it's satisfied...

The equation is satisfied when $q_p \approx 0.6 \text{ mmol/h}$ & $q_p \approx 2 \text{ mmol/g}$

$$\text{Then, } \% \text{ removal} = \frac{(2 \text{ mmol/g}) 5 \text{ g}}{10.638 \text{ mmol}} \times 100\% = \boxed{95\% = \% \text{ removal}}$$

③ a) First, calculate the air flow rate in mol/min (assume ideal gas behavior):

$$\frac{1.25 \text{ m}^3}{\text{min}} \cdot \frac{273 \text{ K}}{423 \text{ K}} \cdot \frac{122 \text{ kPa}}{101.3 \text{ kPa}} \cdot \frac{1 \text{ mol}}{22.4 \text{ L (STP)}} \cdot \frac{10^3 \text{ L}}{1 \text{ m}^3} = 43.37 \text{ mol/min}$$

$$\text{Then, } \Delta \dot{H} + \Delta \dot{E}_k + \Delta \dot{E}_p = \dot{Q} - \dot{W}_s \quad \text{but, } \Delta \dot{E}_k = \Delta \dot{E}_p = 0 \text{ (given)}$$

$$\& \Delta \dot{W}_s = 0 \text{ (no moving parts)}$$

$$\therefore \dot{Q} = \Delta \dot{H} = \dot{n} \Delta \hat{H} = \frac{43.37 \text{ mol}}{\text{min}} \cdot \frac{1 \text{ min}}{60 \text{ s}} \cdot \frac{3640 \text{ J}}{\text{mol}} \cdot \frac{\text{kW}}{10^3 \text{ J/s}}$$

$$\boxed{\dot{Q} = 2.63 \text{ kW}}$$

b) More information would be needed. $\Delta \dot{E}_k$ depends on the cross-sectional areas of the inlet & outlet pipes, so the diameters of the inlet & outlet pipes (or their cross-sectional areas) would be needed.

(5)

(4) Murphy, 6.26

a) Kinetic Energy Change:

$$\begin{aligned}\Delta E_k &= \frac{m}{2} (v_2^2 - v_1^2) = \frac{1 \text{ mol} \cdot 18 \text{ g/mol}}{2} ((100 \text{ km/hr})^2 - (0 \text{ km/hr})^2) \\ &= 9000 \text{ g} \cdot \text{km}^2/\text{hr}^2 \cdot \frac{\text{kg}}{1000 \text{ g}} \cdot \frac{10^6 \text{ m}^2}{\text{km}^2} \cdot \frac{\text{hr}^2}{3600^2 \text{ s}^2} \cdot \frac{\text{J}}{\text{kg} \cdot \text{m}^2/\text{s}^2} \\ &= \boxed{6.9 \text{ J} = \Delta E_k}\end{aligned}$$

b) Potential Energy Change:

$$\begin{aligned}\Delta E_p &= mg(h_2 - h_1) = 1 \text{ mol} \cdot 18 \text{ g/mol} (9.8066 \text{ m/s}^2) (100 \text{ m} - 0 \text{ m}) \\ &= 17651 \text{ g} \cdot \text{m}^2/\text{s}^2 \cdot \frac{\text{kg}}{1000 \text{ g}} \cdot \frac{\text{J}}{\text{kg} \cdot \text{m}^2/\text{s}^2} \\ &= \boxed{17.6 \text{ J} = \Delta E_p}\end{aligned}$$

c) Internal Energy Change due to Pressure Rise
(use steam tables)

$$\begin{aligned}\Delta U &= m \Delta \hat{U} = 18 \text{ g} \cdot (2506.2 - 2515.5) \text{ J/g} \\ &= \boxed{-167 \text{ J} = \Delta U}\end{aligned}$$

⑥

d) Internal Energy Change due to Temperature Rise:
(use liquid H_2O heat capacity)

$$\Delta U = n C_p \Delta T = 1 \text{ mol} \cdot 75.4 \text{ J/mol} \cdot ^\circ\text{C} (100 - 0) ^\circ\text{C}$$

$$= \boxed{7540 \text{ J} = \Delta U}$$

e) Internal Energy Change due to Species Decomposition
(reverse of formation)

$$\Delta \hat{U}_r \equiv \Delta \hat{H}_r - RT \Delta n_r = 285.84 \text{ kJ/mol} \cdot \frac{1000 \text{ J}}{\text{kJ}}$$

$$- 8.3144 \text{ J/mol} \cdot \text{K} \cdot 298 \text{ K} (1 \text{ mol})$$

$$= 283362 \text{ J/mol}$$

$$\Delta U_r = n \Delta \hat{U}_r = 1 \text{ mol} (283362 \text{ J/mol})$$

$$= \boxed{283,362 \text{ J} = \Delta U_r}$$

f) Energy Change due to Nuclear Reaction:

$$\Delta E = \Delta m \cdot c^2 = 18 \text{ g} (2.99792 \times 10^8 \text{ m/s})^2 \cdot \frac{\text{kg}}{1000 \text{ g}} \cdot \frac{\text{J}}{\text{kg} \cdot \text{m}^2/\text{s}^2}$$

$$\boxed{\Delta E = 1.6 \times 10^{15} \text{ J (released)}}$$